(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	**B.Tech.**	Semester	**Fall Semester 2025-26**
Course Code	**BCSE204L**	Faculty Name	**Prof. Sarita Kumari**
Course Title	**Design and Analysis of Algorithms**	Slot	**G1+TG1**
		Class Nbr	**CH2025260102083**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Apply the mathematical tools to analyze and derive the running time of the algorithms
CO2: Demonstrate the major algorithm design paradigms.
CO3: Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
CO4: Articulating Randomized Algorithms.
CO5: Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.

Section - I

Answer all Questions (4 × 10 Marks)

01. During a culinary mystery challenge, Chef Lina hides the recipe for her famous dish. The dish is prepared using N distinct ingredients, each having a specific flavor intensity value (a non-decreasing sequence of positive integers). She challenges Chef Ravi to reconstruct the list of flavor intensities using only the pairwise combined flavor sums of the ingredients. Both chefs know that when any two ingredients (including repetitions) are mixed, their combined intensity equals the sum of their flavor values. Formally, if the sequence of flavor intensities is A = {A[1], A[2], … , A[N]}, then Chef Lina provides Chef Ravi with a shuffled set containing all possible pairwise sums (with repetition) of the form: S={ A[i] + A[j] | 1≤ i ≤ j ≤ N }. The values may appear in any random order. Chef Ravi's task is to reconstruct the original sequence A using a brute-force elimination approach that systematically removes combinations involving already-identified ingredients.

Example:

Let N = 3 and K = 2.

The sequence is A = {A[1], A[2], A[3]}.

All possible 2-sums (with repetition) are: {A[1] + A[1], A[1] + A[2], A[1] + A[3], A[2] + A[2], A[2] + A[3], A[3] + A[3]}.

If the given set of sums is S = {4, 5, 6, 6, 7, 8}, Chef Ravi must reconstruct the original sequence.

Rubrics: [Logic: 3 Marks, Illustration: 3 Marks, Pseudocode: 2 Marks, Time Complexity: 2 Marks]

[10] (CO1/K5)

02. In robotics and genome analysis, strings may "wrap around." Given T = "ABCDEF" and P = "DEFA", determine whether P occurs in T cyclically (i.e., treat T as circular).

a. Modify an existing algorithm to handle circular string matching efficiently. **(5 Marks)**

b. Apply your approach to the above example. **(3 Marks)**

c. Analyze how your modification affects the algorithm's time complexity. **(2 Marks)**

Hint: Think about concatenating T with itself (T+T) and using standard matching.

[10] (CO3/K4)

03. The Smart City Planning Division is designing a green recreation park. They have surveyed eight landmark points (x, y) representing corners of benches, light poles, and small fountains. The design team must create a fencing layout that encloses all these landmarks using the smallest convex polygon possible. The landmark coordinates are:

Landmark	X	Y
A	1	2
B	3	5
C	4	4
D	6	2
E	7	6
F	8	3
G	2	7
H	5	1

a. Identify the starting point (pivot) for Graham's Scan. Sort all other points based on their polar angle with respect to the pivot. **(1 Mark)**

b. Use orientation (cross product) to determine whether to move forward or backtrack while constructing the convex hull. **(2 Marks)**

c. Illustrate each step of stack operations (push/pop) during the algorithm. **(3 Marks)**

d. Draw the final convex polygon (convex hull) showing the boundary of the park. **(2 Marks)**

e. Compare the time complexity of Graham's Scan and Jarvis' March. Discuss a real-world condition (e.g., number of points, collinearity, dataset size) where one method would be preferred over the other. **(2 Marks)**

[10] (CO3/K4)

04. A cybersecurity company is developing a secure key allocation system where each key has a numeric weight representing its security level. The system needs to determine whether a specific total security value can be achieved by selecting a subset of available keys. Formulate this as a Subset Sum problem, and discuss the computational complexity of solving it. Also, explain whether this problem belongs to the NP class or NP-complete.

[10] (CO5/K4)

Section - II

Answer all Questions (4 × 15 Marks)

05. You are given an array of n integers representing the daily stock prices of a company and an integer x, representing a fixed number of consecutive days.
Design a pseudocode using the Divide and Conquer technique to determine the maximum profit that can be obtained by buying on the first day and selling on the last day of any contiguous subarray of length x.
Example: array = [7, 1, 5, 3, 6, 4], x=3
All contiguous subarrays of length x=3:
[7, 1, 5] → buy at 1, sell at 5 → profit = 4
[1, 5, 3] → buy at 1, sell at 5 → profit = 4
[5, 3, 6] → buy at 3, sell at 6 → profit = 3
[3, 6, 4] → buy at 3, sell at 6 → profit = 3
Expected Output: Maximum profit = 4.
Rubrics: [Logic: 3 Marks, Illustration: 5 Marks, Pseudocode: 5 Marks, Time Complexity: 2 Marks]

[15] (CO2/K3)

06. You are given a chain of 7 matrices $\langle A_1, A_2, A_3, A_4, A_5, A_6, A_7 \rangle$, where each matrix A_i has dimension (p[i-1] × p[i]). You want to find the best way to parenthesize these matrices to minimize the total number of scalar multiplications required to compute the product. Consider the constraint: The two middle matrices, A_3 and A_4 must be multiplied together first and treated as a single block. That is, ($A_3 \times A_4$) must stay together and cannot be split during parenthesization. Your task is to modify the Matrix Chain Multiplication (MCM) dynamic programming algorithm to handle this "fixed block" constraint. Compute the optimal parenthesization of $A_1A_2A_3A_4A_5A_6A_7$ respecting the constraint and find the minimum number of scalar multiplications required. Given dimensions p = [5, 10, 3, 12, 4, 6, 8, 2] where $A_1 = 5\times10$, $A_2 = 10\times3$, $A_3 = 3\times12$, $A_4 = 12\times4$, $A_5 = 4\times6$, $A_6 = 6\times8$, $A_7 = 8\times2$.
Rubrics: [Logic: 3 Marks, Illustration: 5 Marks, Pseudocode: 5 Marks, Time Complexity: 2 Marks]

[15] (CO2/K5)

07. A chemist has a set of N compounds, each associated with an energy change value (in integer units). When a compound is added to a solution, it either releases energy (represented as a positive integer) or absorbs energy (a negative integer). The chemist wants to determine whether there exists a subset of these compounds that can be mixed to achieve a net-zero energy change, i.e., the sum of selected values equals 0. The reaction is only considered valid if it uses at least one exothermic compound (positive value) and at least one endothermic compound (negative value). Design a backtracking algorithm for the problem.
Rubrics: [Logic: 4 Marks, Illustration: 4 Marks, Pseudocode: 4 Marks, Time Complexity: 3 Marks]

[15] (CO2/K4)

08. You are designing a communication network between several cities. The cities are represented as nodes in a weighted directed graph G = (V, E), where the weight w(i, j) represents the communication delay (in milliseconds) from city i to city j. If there is no direct connection between two cities, then $w(i, j) = \infty$.
You are required to compute the shortest communication delay between every pair of cities.

a. Explain why Dijkstra's algorithm is not efficient for solving this problem when the number of cities is large, and why Floyd–Warshall is better suited for finding the shortest paths between all pairs of cities. **(2 Marks)**

b. Show an example of a directed weighted graph with 4 cities (nodes) and at least one negative edge weight (but no negative cycle). Explain how Floyd–Warshall handles this situation. **(2 Marks)**

c. Derive the dynamic programming recurrence relation used in the Floyd–Warshall algorithm. Clearly define the subproblem and the meaning of each DP entry $D^{(k)}[i][j]$. **(3 Marks)**

d. In the iterative implementation of Floyd–Warshall: **(4 Marks)**

- What does the matrix $D^{(k)}$ represent at the kth step?
- What should the initial matrix $D^{(0)}$ contain?
- How does the algorithm progressively update the values in the matrix?

e. Write pseudocode for the Floyd–Warshall algorithm that computes the shortest path matrix between all pairs of nodes. Extend the pseudocode to also reconstruct the actual path (sequence of cities) between any two given cities. **(4 Marks)**

[15] (CO3/K5)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

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